Charm Industrial Isometric Standard Project Verification Report

Client	Isometric
Company / GHG Removal Supplier	Charm Industrial
Facility or Project Name	Charm Kansas Bio-Oil Geologic Storage
Facility Address	**Bio-oil Production:** **AE Cote-Nord Canada Bioenergy Inc (AECN)** Port Cartier, Quebec, Canada **Charm Miniforge** Fort Lupton, CO, USA **Bio-Oil Pre-Processing:** **Charm Industrial** El Dorado, KS, USA **Bio-oil Injection:** **Vaulted Deep** Hutchinson, KS, USA
Greenhouse Gas Statement ID	Isometric gss_1JQC61HS91S06E2C
Net Greenhouse Gas Removals	592.66 tonnes
Removal Method	Bio-Oil Geological Storage
Removal Period	Dec 7, 2024 – Jan 31, 2025
Verification Opinion	Fully meets criteria
Verifier/Validation and Verification Body	Tim Hansen Lily Schacht 350Solutions, Inc.

ISSUED: APRIL 16, 2025

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ISOMETRIC STANDARD VERIFICATION REPORT		
Company: Charm Industrial	Company Contact: *Max Lavine	Validation and Verification Team: *Tim Hansen, PE Lily Schacht
Removal Method: Bio-Oil Geological Storage		
Report Date: April 16, 2025		
Document No: 350VR-CHARM-ISO2504-RPE31JAN25		
Rev: v1.2		

** primary contact/lead author*

1. KEY PROJECT DATA

TABLE 1. CHARM TECHNOLOGY VERIFICATION SUMMARY

Title	Charm Kansas Bio-Oil Geologic Storage
Location	Bio-oil Production: AE Cote-Nord Canada Bioenergy Inc (AECN) @ Produits Forestiers Arbec S.E.N.C., 175 Bd du Portage des Mousses Port-Cartier, QC G0W 1X0, Canada 50.025323, -66.818910 Charm Industrial Miniforge 2360 S Rollie Ave Fort Lupton, CO, USA 80621 40.076570, -104.804680 Bio-Oil Pre-Processing: Charm Industrial 838 SW Purity Springs Rd El Dorado, KS 67042 37.813906, -96.914982 Bio-oil Injection: Vaulted Deep 7513 KS-14 Hutchinson, KS 67501 37.966035, -97.947845
Reporting Period	December 7, 2024 – January 31, 2025
Greenhouse Gas Statement ID	Isometric ggs_1JQC60KVR1S0JS6Q (Statement ID) Isometric gss_1JQC61HS91S06E2C (Version ID)
Project Proponents	CDR Supplier: Charm Industrial Feedstock Supplier: Produits Forestiers Arbec S.E.N.C (Quebec mill waste supplier); Altitude Forestry (CO forestry fire prevention waste supplier) Geologic Storage Management: Vaulted Deep Regulatory Agency (Injections / Storage): Kansas Department of Health and Environment (KDHE)
Legal Ownership of the Credits	Charm Industrial
Removals in Reporting Period	592.66 tonnes
Removal Method	Bio-Oil Geological Storage in Salt Caverns

Verification Report Date	April 16, 2025
Verification Type & Criteria	Isometric Standard Verification for ongoing deliveries of carbon removal credits by Charm Industrial in accordance with the documents used during project validation: • Isometric Standard v1.4 • Bio-Oil Geological Storage Protocol v1.1 • Biomass Feedstock Accounting Module v1.2.1 • Biomass or Bio-oil Storage in Salt Caverns v1.1 • Embodied Emissions Accounting v 1.0 • Energy Use Accounting v 1.2 • Transportation Emissions Accounting v1.1

1.1. VERIFICATION SUMMARY

350Solutions was contracted to perform a project verification of the greenhouse gas (GHG) statement of Charm Industrial's bio-oil geological storage process for reporting period including December 7, 2024 – January 31, 2025. The validation report for the Charm Kansas Bio-oil Geologic Storage project was completed on June 25, 2024, by Futurepast, which focused on the use of third-party produced bio-oils and was issued by Isometric for the Crediting Period March 1, 2024 – February 28, 2029. An additional validation of the Charm Kansas Bio-oil Geologic Storage project for use of bio-oil produced by Charm was completed on October 18, 2024 for the same crediting period. These validations remain valid for this verification reporting period. Further details and findings can be found in the validation report (https://registry.isometric.com/project/prj_1HZSSWBQM1S08H83#project-documentation).

350Solutions affirms that Charm has the appropriate equipment, procedures, and protocols in place and documented to quantify GHG removal through the injection of bio-oil into salt caverns via permitted Class V wells in accordance with the requirements of the applicable Isometric standards, protocols and modules.

The carbon removals for the reporting period are provided in Table 2. These values have been verified by 350Solutions.

TABLE 2. VERIFIED TOTAL REMOVALS BY CHARM INDUSTRIAL FOR REPORTING PERIOD ENDING JANUARY 31, 2025

Removal ID	Supplier reference ID	Date removal started	Date removal completed	Discounted CO_2e net removed, rounded (kg)
rmv_1JPTQFN3C1S0KYX8	755	2024-12-24	2024-12-24	9950
rmv_1JPWY7BHR1S0Z2E7	765	2025-01-24	2025-01-24	49920
rmv_1JPWYJAAQ1S0S2T5	759	2024-12-31	2024-12-31	24220
rmv_1JPWYS3QR1S0VSYD	760	2025-01-08	2025-01-08	14790
rmv_1JPWYXS4K1S0QXC7	761	2025-01-13	2025-01-13	49710
rmv_1JPWZ3N221S0D7V6	762	2025-01-15	2025-01-15	35300
rmv_1JPWZFXN41S05460	764	2025-01-23	2025-01-23	52100
rmv_1JPWZMJ8G1S0FV2E	766	2025-01-25	2025-01-25	49760
rmv_1JPWZXVYB1S0YP44	768	2025-01-30	2025-01-30	50050
rmv_1JPX09P081S0FM7P	763	2025-01-18	2025-01-18	47120
rmv_1JPX27CEM1S0FVTR	767	2025-01-31	2025-01-31	50630

rmv_1JPX3YAWX1S0Q176	754	2024-12-19	2024-12-19	11070
rmv_1JPX42BRN1S040T2	756	2024-12-27	2024-12-27	48740
rmv_1JPX47F8X1S0JK9E	757	2025-01-02	2025-01-02	49560
rmv_1JPX4AFH41S0PJ69	758	2025-01-04	2025-01-04	49740
TOTAL REMOVALS				**592660 kg (592.66 tonnes)**

The quantity of 592.66 tonnes of net carbon removals has been verified with a reasonable level of assurance and are eligible for crediting under the referenced Isometric Standard and Protocol.

350Solutions Inc. declares that it is an impartial verification body, approved by Isometric for verifications in the required technical area, free from any conflicts of interest, capable, and qualified to complete this verification for the current reporting period according to the *Isometric Standard V1.4, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2.1*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*.

Verifier Information		
Verification Body	**Verifier**	**Verification ID No.**
350Solutions, Inc.	Tim Hansen	VR-Charm-ISO2504-RPE31JAN2025

Timothy A. Hansen
Digitally signed by Timothy A. Hansen
Date: 2025.04.16 16:07:16 -04'00'
Signed: Tim Hansen (Verifier)

Lily Schacht
Digitally signed by Lily Schacht
Date: 2025.04.16 16:11:30 -04'00'
Lily Schacht (QA)

2. VERIFICATION DETAILS

2.1. VERIFICATION OBJECTIVES

A series of verification activities was conducted by 350Solutions to independently verify the claimed removals of the GHG statement submitted by Charm Industrial for operations conducted at the Kansas Bio-oil Geologic Storage facility for the time-period December 7, 2024 – January 31, 2025. The verification was conducted following the specifications of *Isometric Standard V1.4, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2*, *Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1,* and *Energy Use Accounting v1.2*. The verification is based on the provision and fair presentation of a complete and accurate GHG Statement by Charm Industrial and Isometric. This is confirmed and verified by 350Solutions based on supporting evidence in accordance with the Isometric criteria (Table 1, Section 3.2).

2.2. LEVEL OF ASSURANCE & MATERIALITY

The verification was conducted with a reasonable level of assurance. Any discrepancies between the claimed removals of the project and the removals estimated by the verification process for the reporting period are less than the materiality threshold of 5%.

2.3. VERIFICATION TEAM DETAILS

350Solutions is an accredited inspection & verification body by ANAB under ISO 17020 for completion of ISO 14034 Technology Verifications. 350Solutions Technical Lead for the Charm verification is Tim Hansen. Mr. Hansen is a registered Professional Engineer with over 20 years of experience in the evaluation and verification of novel cleantech and carbon technologies. Mr. Hansen has led verifications of carbon removal technologies for multiple registries, including Isometric, and has experience in the development and testing of biomass conversion technologies, including gasification and pyrolysis. The verification report was written by Lily Schacht, a Senior Carbon Removal Verification Engineer, with 4 years of experience developing and verifying low-carbon and carbon removal technologies. Complete verifier qualifications are attached in Appendix 3.

2.4. AUDIT PROCESS FOLLOWED

The specific verification activities conducted are summarized in Table 3. The list of all files that were reviewed is provided in Appendix 1. The list of findings from the verification can be found in Appendix 2. Audit results are documented using 350's Compliance Checklist, which includes all requirements from the Isometric standard, Bio-oil geologic Storage protocol, and referenced modules.

TABLE 3. VERIFICATION ACTIVITIES

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
March 31 – April 11, 2025	Review of reporting period modifications to PDD	- Review of updated PDD and GHG Statement	V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx (and V 1.8 – originally submitted for this RP) Dec_Jan Removals Isometric GHG Statement Report v1.0.docx Dec_Jan Verification Guidance

Date(s)	Verification Activity	Verification Tasks	Documents Reviewed
			ggs_1JQC60KVR1S0JS6Q / gss_1JQC61HS91S06E2C
March 31 – April 11, 2025	Review of current Validation and GHG Statement Applicability	- Review of Validation Report - Review of prior Verification Report - Identification of major and minor process steps, measurements, and calculations - Review of Standards/Protocols/Modules required for Verification - Generation of a checklist of verification issues, recommendations, and opportunities for improvement	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1 Charm Verification Report 2024-07-26 Charm_ISO2405_VRFINALV1.2 CharmBioOil_ISO2405_VR-FINAL_v1.3_10-18-2024 V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx (and V 1.8 – originally submitted for this RP) Dec_Jan Removals Isometric GHG Statement Report v1.0.docx Dec_Jan Verification Guidance Isometric Standard V1.4, Bio-oil Geological Storage Protocol v1.1, Biomass Feedstock Accounting Module V1.2, Bio-oil Storage in Salt Caverns Module v1.1, Embodied Emissions Accounting v1.0, Transportation Emissions Accounting v1.1, and Energy Use Accounting v1.2
March 31 – April 11, 2025	Verification of Removals	- Verify changes to the system - Review of measurements and calculations - Review of durability/permit requirement monitoring data - Review of GHG statement - Identification of initial findings and follow-up request for information - Write verification report	File List provided in Appendix 1. Initial Findings in Appendix 2.

3. TECHNICAL ASPECTS

3.1. PROJECT DETAILS

Charm obtained bio-oil from two sources during this reporting period. Charm obtains bio-oil from a producer in Quebec, Canada (AECN) who uses an Ensyn/ Honeywell UOP pyrolysis system to generate bio-oil from waste wood produced at a neighboring mill. In addition, Charm now produces bio-oil at its pyrolysis facility in Fort Lupton, CO. Production of bio-oil for sequestration and carbon removal credit generation began in August of 2024, with the initial installation of 4 pyrolyzers, with plans for up to 18 pyrolyzers total to be installed. Charm's pyrolyzers utilize wood chips, placed into a high temperature pressurized reactor in a low oxygen environment to reduce the wood chips into a combination of products, including syngas and light hydrocarbon gases, pyrolysis oils, biochar, and wood vinegar. All hot pyrolysis gases from the reactor are quenched with water to produce the oil and wood vinegar products. Excess gases are routed to a combustion device for emissions control, with the heat produced from the combustion of excess gas utilized for drying the biomass from 40%-50% moisture content to 10% moisture content before introduction into the pyrolyzer.

Bio-oil from both facilities is shipped by tanker truck or rail car to a Charm facility in El Dorado, Kansas. At this facility, Charm pretreats the bio-oil through the addition and sparging of waste brine (salt) and liquid caustic soda (LCS) to adjust the pH of the bio-oil and the salt content or conductivity to meet the requirements of the injection facility permit for storage in salt caverns as specified by the Kansas

Department of Health and Environment (KDHE) and injection partner Vaulted Deep's Class V injection permit. Following treatment, samples are taken for CHN analysis (carbon content), total acid number (TAN), and density testing and the oil is transported to the Vaulted Deep injection facility.

Once successfully pre-treated to ensure EPA Underground Injection Control (UIC)/KDHE compliance, including measurement of pH and conductivity both within allowable permitted ranges, the oil is trucked to Vaulted Deep's Great Plains Facility in Hutchinson, KS for injection into an underground salt cavern. At that point, the oil is offloaded and Vaulted takes custody of the product and responsibility for emplacing it into the storage cavern. The waste is not stored on site, instead it is processed and injected on the same day as it arrives on site. Therefore, emissions associated with waste storage at the facility are avoided. Vaulted is also contracted to monitor the emplacement cavern and provide Charm with MRV data from their monitoring equipment. The details of Vaulted's injection infrastructure and monitoring equipment are detailed in their own PDD, which has been previously validated [1].

3.2. PROTOCOLS AND MODULES APPLIED

The verification was conducted following the specifications of the following standards, protocols, and modules:

- Isometric Standard v1.4 [2]
- Bio-Oil Geological Storage Protocol v1.1 [3]
- Biomass Feedstock Accounting Module v1.2 [4]
- Biomass or Bio-oil Storage in Salt Caverns v1.1 [5]
- Embodied Emissions Accounting v 1.0 [6]
- Energy Use Accounting v 1.2 [7]
- Transportation Emissions Accounting v 1.1 [8]

Transportation Emissions Accounting Module v1.0 was used in the initial validation and previous verifications of this project. The module was updated (V1.1) to allow for the use of Book and Claim Units (BCUs) to substitute for some, or all, or project fuel usage.

3.3. APPLICABILITY OF APPLIED PROTOCOL AND MODULES

The Charm project takes mixed woody waste or forestry thinnings from wildfire prevention activities, converts it to bio-oil, which they pretreat and inject into salt caverns. The biomass feedstocks were assessed previously under the eligibility criteria of the Isometric Biomass Feedstock Module and were found to be eligible. As the biomass feedstocks have been deemed eligible biomass feedstocks, the storage of biomass conversion products (bio-oil) in underground salt caverns makes both the Bio-oil Geological Storage Protocol and Biomass/Bio-oil Storage in Salt Caverns Module applicable to this project. All other modules for energy, transportation, and embodied emissions accounting are generally applicable to this project.

3.4. PROJECT BOUNDARY

The boundaries of the project include the supply of wood waste to the biomass processing sites (AECN and Charm Fort Lupton), the conversion of the biomass to bio-oil via AECN's pyrolysis process and Charm's pyrolysis process, transport of the bio-oil to Charm for preprocessing at its El Dorado, Kansas facility, completion of pretreatment, shipment to the Vaulted Deep Great Plains facility, and injection of the bio-oil in the salt caverns. Monitoring and required analysis of the bio-oil and the storage formation are also included in the project boundary.

3.5. CHANGES FROM PREVIOUS VERIFICATIONS

Compared to prior verifications, Charm has made several modifications to the process which have been identified in the GHG statement and supporting documentation. The following modifications in project activities were made, in addition to updates to delivery data and monitoring data for the reporting period:

- This reporting period included the first injections of bio-oil produced by Charm at its Fort Lupton, CO facility. This process was validated in October, 2024. As a result of these first injections, several new data files and supporting evidence files were provided for verification, including:
    - Calculation of the lifecycle emissions and carbon intensity of the Charm pyrolysis process;
    - Measurements and analysis of Charm bio-oil product;
    - Supporting evidence for all components of Charm bio-oil production including shipping bills of lading, measurements of bio-oil production mass and carbon content, documentation of energy use by the facility, and other relevant information;
    - Evidence supporting forward action requests from the validation;
    - Uncertainty and sensitivity analyses for Charm bio-oil production;
- Disposal of bio-oil was required at three points during this reporting period:
    - Lots 200062 and 200063 were scrapped due to operational issues in pre-processing. Oil has been stored on site and is shipped in totes for disposal over time. All emissions associated with production, disposal, and transport of these bio-oil batches are included in the net removals calculation.
    - Lot 200065 was injected at the Kansas storage site, however, carbon content samples were not properly collected and analyzed. As a result, Charm has not claimed credit for the removals. However, all emissions associated with production of this bio-oil batch were included in the net removals calculation.
    - Bio-oil in one railcar was disposed of as the result of a reaction with the lining of the railcar. All emissions associated with production, disposal, and transport of these bio-oil batches are included in the net removals calculation.
- In addition to book and claim units being used to offset the emissions associated with fossil diesel usage for transportation with emissions associated with biodiesel usage, as performed in the prior reporting period, Charm has introduced the use of RECs to offset the emissions from electricity usage at its Colorado pyrolysis facility and the Kansas pre-processing facility. RECs were reviewed for compliance with the Energy Use Accounting module and for proper emissions accounting. Charm has included the embodied emissions of the REC source (wind in KS, solar in CO) in the net removal calculations.
- All new equipment and materials have been inventoried and embodied emissions accounted for, which are allocated to removals appropriately and in accordance with the Isometric protocol requirements.

4. CREDITING DETAILS

4.1. PROJECT REPORTING PERIOD

This verification considers injections of bio-oil by Charm during the period December 7, 2024 to January 31, 2025. The reporting period is based on the actual injection dates, with bio-oil production,

transportation, monitoring, and related activities potentially occurring outside of this reporting period, but properly accounted for and allocated to the bio-oil injected within the reporting period.

4.2. CREDITING PERIOD

The 5-year crediting period for the Charm Kansas Bio-oil Geologic Sequestration Project began March 1, 2024 and continues for five years, ending February 28, 2029. All claimed removals in this reporting period fall within the current crediting period.

5. GHG STATEMENT VERIFICATION

5.1. ACCURACY OF THE STATEMENT

The values represented in the GHG statement have been evaluated, cross-referenced with the presented evidence, and confirmed. Confirmation of the GHG statement has been based on the following efforts:

- Duplication of calculations of total bio-oil and CO_2 equivalent injected for removals rmv_1JPTQFN3C1S0KYX8 (755), rmv_1JPX47F8X1S0JK9E (757), and rmv_1JPX3YAWX1S0Q176 (754), including:
    - Calculation of mass of bio-oil delivered and injected including tracing to raw data (weigh scale tickets, carbon content, addition of consumables (salt and LCS)
    - Review and calculation of average carbon content from raw data (lab reports)
    - Confirmation of transportation distances for tanker and rail car deliveries from AECN to El Dorado, KS
    - Confirmation of weigh scale raw data for deliveries
    - Re-calculation of transportation associated emissions for such transport.
    - Confirmation of raw data for fuel usage at Charm pre-treatment facility
    - Confirmation of electricity usage at Charm pretreatment facility and Vaulted Deep injection facility from raw data
    - Confirmation of proper use and accounting of BCUs and RECs to offset diesel and electricity usage, respectively
    - Confirmation of emission factors for RECs embodied emissions
    - Review of Charm pyrolysis process emissions, supporting evidence, and calculations
- Review of monitoring data as required for salt cavern injection / emplacement and post injection monitoring to ensure ongoing compliance with Class V permit.

As with the previous verification, some minor discrepancies were noted due to rounding errors both internally at Charm and in communication between Charm's and Isometric's data platforms. The estimated impact of the total discrepancies is not a material issue.

5.2. VERIFICATION OPINION

350Solutions has reviewed the documentation of the technology, the instrumentation, the procedures, performance and collected data and has found that the data presented in the greenhouse gas removal statement indicates that the claims fall within the 5% materiality threshold for greenhouse gas emissions and removals as required by the Isometric Standard:

☒ ***Meets the requirements of the Isometric Standard V1.4, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Meets the requirements with minor revisions of Isometric Standard V1.4, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

☐ ***Does Not Meet the requirements of the Isometric Standard V1.4, Biomass Geological Storage Protocol V1.1, Biomass Feedstock Accounting Module V1.2, and Biomass or Bio-oil Storage in Salt Caverns Module v1.1 for Verification***

5.3. VERIFICATION RESULTS SUMMARY

A summary of verification results and compliance with relevant requirements of the *Isometric Standard, Biomass Geological Storage Protocol, Biomass Feedstock Accounting Module*, and *Biomass or Bio-oil Storage in Salt Caverns Module* are summarized below.

TABLE 4. VERIFICATION RESULTS

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
2.1.	*Validated against approved Protocols*	N	Acceptable – No change since validation
2.5.1	*Boundaries*	N	Acceptable – No change since validation
2.5.2	*Baseline*	N	Acceptable – No change since validation
2.5.3	*Additionality - Financial*	N	Acceptable – No change since validation
2.5.3	*Additionality - Environmental*	N	Acceptable – No change since validation
2.5.3	*Additionality - Regulatory*	N	Acceptable – No change since validation
2.5.4	*Leakage*	N	Acceptable – No change since validation
2.5.5	*Default Emission Factors, Proxies, and Models*	Y	Acceptable – Charm-produced bio-oil lifecycle assessment and emissions calculations have been provided for the first time. After findings were addressed, emission factors, calculations, and uncertainty analyses comply with protocol requirements. With the addition of RECs usage, additional emission factors and calculations were used for electricity (REC) emissions. All calculations and factors comply with protocol requirements. No additional significant changes were made since prior validations.
2.5.11	*Life Cycle Assessment (LCA) Policies*	Y	
2.5.6	*Common Calculation Factors*	Y	
2.5.7	*Uncertainty*	Y	
2.5.8	*Durability and Monitoring*	N	Acceptable – No change since validation
2.5.9	*Risk of Reversal*	N	Acceptable – No change since validation
2.5.10 ***3.7***	*Environmental and Socioeconomic Impacts*	N	Acceptable – No change since validation.
3.1	*Ownership*	N	Acceptable – No change since validation

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
3.2	*Documentation*	Y	Acceptable – Charm provided significant new documentation in support of the Charm bio-oil production process. No other changes since validation
3.3	*Eligibility*	Y (GHG Statement Updated)	Acceptable – Project registered with Isometric and reporting period GHG Statement properly submitted to Isometric dMRV platform.
3.4	*Project Crediting*	Y (GHG Statement Updated)	Acceptable – New reporting period GHG Statement is complete and all removals have been verified. Reporting period is within current validation crediting period.
3.5	*Stakeholder Input Process*	N	Acceptable – No change since validation. Charm did provide updated community engagement documentation in support of the operations at the Charm Fort Lupton pyrolysis facility.
3.6	*Regulatory Compliance*	Y (Required Monitoring Updated)	Acceptable – Maintenance of well operation permits meets requirements and ongoing permit reporting requirements completed.
Biomass Feedstock Accounting Module			
2.1	*Eligibility Criteria for Market Leakage*	N	Acceptable – No change since validation
2.2	*Counterfactual Storage Eligibility*	N	Acceptable – No change since validation
2.3	*Dedicated Energy Feedstock Eligibility*	N	Acceptable – No change since validation
3.0	*Counterfactual Definition*	N	Acceptable – No change since validation
3.1	*Counterfactual Energy Definition*	N	Acceptable – No change since validation
3.2	*Counterfactual Replacement Definition*	N	Acceptable – No change since validation
Biomass or Bio-oil Storage in Salt Caverns Module			
1.0, 2.2	*Salt Cavern Characterization – Cavern Dimensions*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Integrity*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Porosity & permeability*	N	Acceptable – No change since validation
1.0, 2.2	*Salt Cavern Characterization – Subsurface Features*	N	Acceptable – No change since validation
1.0	*Salt Cavern Characterization – Temperature, pH, conductivity/Cl- conc.*	Y (Required Monitoring Updated)	Acceptable - All required parameters measured and are within the required limits.
1.0	*Salt Cavern Characterization – Brine TOC*	Y (Required Monitoring Updated)	Acceptable – Brine TOC measured prior to emplacement, as required by Class V permit. Confirmed via updated TOC data.

Isometric Standard Section Ref.	Verification Topic	Changed since Project Validation? (Y/N)	Final Findings
1.0	*Salt Cavern Characterization – $d^{13}C$ analysis (where applicable)*	NA	Acceptable – No change since validation
1.0, 3.1.3	*Salt Cavern Characterization – Surface Subsistence*	NA	Acceptable - No change since validation
1.0	*Salt Cavern Characterization – Displaced Brine Injection location*	NA	Acceptable - No change since validation
1.0, 2.3	*Emplacement Infrastructure*	N	Acceptable – No change since validation
1.0	*Emplacement Monitoring*	Y (Required Monitoring Updated)	Acceptable – monitoring completed and updated as required by Class V well permit
1.0, 3.2	*Well Closure and Post Closure Requirements*	N	Acceptable – No change since validation
2.1	*Permit for Operation*	N	Acceptable – No change since validation
3.1.1	*Injection Pressure*	Y (Required Monitoring Updated)	Acceptable – Pressure is continuously monitored and reported to KDHE monthly. Updates provided as required.
3.1.1	*Volume monitoring*	Y (Required Monitoring Updated)	Acceptable – Monitoring of injected material measured daily and reported monthly, confirmed by sonar quarterly. Updates provided as required.
3.1.1	*Injectate monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.1	*Sampling Frequency*	N	Acceptable – No change since validation. All sampling data provided and updated as required
3.1.1	*Safety Monitoring*	N	Acceptable – No change since validation
3.1.2	*Mechanical Integrity*	N	Acceptable – No change since validation
3.1.3	*Well Pressure Monitoring*	Y (Required Monitoring Updated)	Acceptable – Well pressure measured continuously with safety interlock. Daily pressure reported to KDHE monthly. Updated monitoring data provided as required.
3.1.3	*Sonar Survey*	N	Acceptable – No change since validation
3.1.3	*Fill Volume*	Y (Required Monitoring Updated)	Acceptable – Cavern volume tracked based upon volume of emplaced material. Estimate volume reported monthly, verified by sonar quarterly. Updated monitoring data provided as required.
3.1.3	*Return Brine Monitoring*	Y (Required Monitoring Updated)	Acceptable – Chloride content (conductivity), pH, and temperature measured continuously, TOC measured once per month, while all values are reported monthly. Updated monitoring data provided as required.
3.1.3	*Volatile Gas Monitoring*	N	Acceptable – No change since validation
3.3	*Reversal Risk*	N	Acceptable – No change since validation
3.3.1	*Attribution of Reversals*	N	Acceptable – No change since validation

5.4. VERIFICATION FINDINGS

Assessment of the greenhouse gas removal statement identified several initial findings (Appendix 2). All findings, primarily associated with missing supporting evidence, have been addressed and closed.

Forward Action requests (FAR) and recommendations are also provided in Appendix 2 for consideration for improvement of future operations and GHG Removal Credit calculations and to ensure tracking of potential issues to be monitored by verifiers in future verifications. Open Forward Action Requests and opportunities for improvement are summarized below for future verification reference:

TABLE 5. FORWARD ACTION REQUESTS AND RECOMMENDATIONS

Type	Finding / Issue	Source	Protocol Reference	Action
Forward Action Request	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing.	Validation_Report_for_Isometric_Charm_Industrial_Bio-Oil_Project_2024-06-25_r1	7.4.3.5	Please provide AECN testing report. Note, not applicable for this reporting period, but should be updated when possible
Forward Action Request	Disposal of bio-oil from batches 200062 and 200063 may continue over several reporting periods	Dec_Jan Verification Guidance.docx	7.4.3	Please provide evidence and accounting of disposal of oil from lots 200062 and 200063 when disposed of in future
Forward Action Request	Biomass feedstock sourcing was confirmed at validation. Because of its critical nature for compliance, we recommend confirming for each verification reporting period if possible, demonstrating no change in feedstock sources since validation		7.2	Please provide confirmation of biomass feedstock sourcing and that it is not different from validated sources (mill wastes) when possible for verifications in the future
Forward Action Request	The planned change to a low swirl burner for Charm pyrolysis gas emissions control was not implemented, but modifications to the existing system were and emission levels re-evaluated for use in future reporting periods (conservatively not used in current RP)	Charm BO Validation FARs.xls	7.4.3.5	Please provide a copy of the October 2024 emissions control burner test when used in future removals accounting for tail gas emissions

6. REVISION HISTORY

Original date of issue: February 20, 2025

Version	Date Issued	Noted Changes
V1.0 (Draft – Internal)	April 11, 2025	N/A Initial Version
V1.1 (Draft – External)	April 11, 2025	
V1.2 (FINAL)	April 16, 2025	Minor edits based on QA and client/supplier review. Correction of dates of removal rmv_1JPWYXS4K1S0QXC7.

References

[1] "Vaulted Deep Great Plains Facility Organic Waste Sequestration: Isometric PDD." Jan. 23, 2025. Accessed: Feb. 19, 2025. [Online]. Available: https://storage.googleapis.com/isometric--production--eu--public-documents/spl_1H43SYY631S0Z8YR/prj_1HHYZFVGW1S044ZY/pjd_1JJSMS9FS1S0MF1M/pdd.pdf

[2] Isometric, "Isometric Standard v1.2.0." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/standard/1.2.0

[3] Isometric, "Bio-oil Geological Storage V1.1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/bio-oil-geological-storage/1.1

[4] Isometric, "Biomass Feedstock Accounting V1.1." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/biomass-feedstock-accounting/1.1

[5] Isometric, "Biomass or Bio-oil Storage in Salt Caverns V1.0.4." Accessed: Feb. 19, 2025. [Online]. Available: https://registry.isometric.com/protocol/salt-cavern-storage/1.0

[6] Isometric, "Embodied Emissions Accounting v1.0.3." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/embodied-emissions/1.0

[7] Isometric, "Energy Use Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/energy-use-accounting/1.1

[8] Isometric, "Transportation Emissions Accounting v1.1." Accessed: Nov. 20, 2024. [Online]. Available: https://registry.isometric.com/protocol/transportation

See Appendix 1 and Table 3 for list of specific files reviewed during the verification.

APPENDIX 1: DOCUMENTS SUBMITTED FOR REVIEW

File Type	File Name
PDD	V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx (and v1.8 – 2025/02/19)
GHG Statement	Dec_Jan Removals Isometric GHG Statement Report v1.0.docx (1)
GHG Statement	Dec_Jan Verification Guidance
GHG Statement Data from Isometric Platform	gss_1JQC61HS91S06E2C
Source Data	
Batch Information / Logs	200062.xlsx
Batch Information / Logs	200063.xlsx
Batch Information / Logs	200065.xlsx
Batch Information / Logs	200066 MRV Ops Notes.xlsx
Batch Information / Logs	200067 MRV Ops Notes.xlsx
Batch Information / Logs	200067 Truck 1.pdf
Batch Information / Logs	200067 Truck 2.pdf
Batch Information / Logs	200068_300001 MRV Ops Notes.xlsx
Batch Information / Logs	200069 MRV Ops Notes.xlsx
Batch Information / Logs	200070 MRV Ops Notes.xlsx
Batch Information / Logs	200072 MRV Ops Notes.xlsx
Batch Information / Logs	200073 MRV Ops Notes.xlsx
Batch Information / Logs	200074 MRV Ops Notes.xlsx
Batch Information / Logs	200071 MRV Ops Notes.xlsx
Batch Information / Logs	300002 MRV Ops Notes.xlsx
Batch Information / Logs	300003 MRV Ops Notes.xlsx
Batch Information / Logs	300004 MRV Ops Notes.xlsx
Batch Information / Logs	300005 MRV Ops Notes.xlsx
Batch Information / Logs	300006 MRV Ops Notes.xlsx
Batch Information / Logs	Batch 200066 - 200068 MRV(CHN + H2O).pdf
Batch Information / Logs	Batch 200066 MRV(CHN + H2O) (1).pdf
Batch Information / Logs	BATCH 200066-68 MRV(TAN + Density) (2).pdf
Batch Information / Logs	BATCH 200067 MRV(CHN + H2O).pdf
Batch Information / Logs	BATCH 200067 MRV(TAN + Density).pdf
Batch Information / Logs	BATCH 200068 MRV(TAN + Density).pdf
Batch Information / Logs	BATCH 200069, 200070, 200071, 300002 #1 MRV (CHN + H2O).pdf
Batch Information / Logs	BATCH 200069, 200070, 200071, 300002 #1 MRV (TAN + Density)).pdf
Batch Information / Logs	Batch 200072 Truck 1 Scale Ticket.pdf
Batch Information / Logs	Batch 200072 Truck 2 Scale Ticket.pdf
Batch Information / Logs	BATCH 3-03 FINAL MRV.pdf
Batch Information / Logs	BATCH 3-03 FINAL SPARGED.pdf
Batch Information / Logs	BATCH 3-05 #1, BATCH 200072 #1, BATCH 200073 #1 TAN + Density.pdf

Batch Information / Logs	BATCH 3-05 #1, BATCH 200072 #1, BATCH 200073 #1, CHN + H2O.pdf
Batch Information / Logs	BATCHES 200074, 200075, 3-06 MRV (CHN + H2O).pdf
Batch Information / Logs	BATCHES 200074, 200075, 3-06 MRV (TAN + Density).pdf
Batch Information / Logs	CHARM INDUSTRIAL - BATCH 3-04 CHN + H2O #1.pdf
Batch Information / Logs	CHARM INDUSTRIAL - BATCH 3-04 TAN + Density #1.pdf
Batch Information / Logs	Charm-Blended Batch C Content.xlsx
Bio-oil sourcing	Docs Charm 1910 CORRECTIONS.pdf
Bio-oil sourcing	Document Camion 1426.pdf
Bio-oil sourcing	Document Camion 1780.pdf
Bio-oil sourcing	Document Camion 1798.pdf
Bio-oil sourcing	Document Truck 1799.pdf
Bio-oil sourcing	Documents Camion 1786 - Billing Documents 1786.pdf
Bio-oil sourcing	Documents Camion 1838.pdf
Bio-oil sourcing	Documents Camion 1840.pdf
Bio-oil sourcing	Documents Camion 1877.pdf
Bio-oil sourcing	documents camion charm 1853.pdf
Bio-oil sourcing	Documents camions 1425.pdf
Bio-oil sourcing	documents charm 1907.pdf
Bio-oil sourcing	documents charm 1913.pdf
Bio-oil sourcing	documents charm 1915.pdf
Bio-oil sourcing	envois de documents 1428.pdf
Charm Bio-Oil Non-Hazardous Classification Testing	2024.11 Charm Industrial Waste Profile .pdf
Charm Bio-Oil Process Emissions	Consolidated Energy Consumption.xlsx
Charm Bio-Oil Process Emissions	Miniforge Biomass Transport Allocation -- PP1 w_ Uncertainty Adj..xlsx
Charm Bio-Oil Process Emissions	Production Period 1 Summary -- Final Submission.xlsx
Community Engagement Docs	Charm Colorado Community Outreach_Aug-Dec 2024.pdf
Diesel Fuel Usage	Dec_Jan Diesel Consumption Calcs.xlsx
Diesel Fuel Usage	Secuirty Oil 120924.pdf
Diesel Fuel Usage	Security Oil 01032025.pdf
Diesel Fuel Usage	Security Oil 01072025.pdf
Diesel Fuel Usage	Security Oil 01142025.pdf
Diesel Fuel Usage	Security Oil 01202025.pdf
Diesel Fuel Usage	Security Oil 01272025.pdf
Diesel Fuel Usage	Security Oil 121224.pdf
Diesel Fuel Usage	Security OIl 12152024.pdf
Diesel Fuel Usage	Security Oil 12162024.pdf
Diesel Fuel Usage	Security Oil 12202024.pdf
Diesel Fuel Usage	Security Oil 12232024.pdf
Diesel Fuel Usage	Security Oil 12302024.pdf
Emission Factor Documentation	CO Grid Emissions Calc Sheet.xlsx
Emission Factor Documentation	Treatment of Food Waste.pdf
Emission Factor Documentation	Treatment of Waste Wood.pdf

GHG Statement	gss_1JQC61HS91S06E2C.xlsx
GHG Statement Report	Dec_Jan Removals Isometric GHG Statement Report v1.0.docx (1).pdf
Injection Well Permit Reporting	ACS 2024 Annual Report - AR-09.pdf
Injection Well Permit Reporting	ACS 2024 Fourth Quarter Report - AR-09.pdf
Injection Well Permit Reporting	AR-09 Displaced Brine 1-16-25.pdf
Injection Well Permit Reporting	AR-09 Displaced Brine 12-18-24.pdf
Injection Well Permit Reporting	AR-09 Displaced Brine 1-2-25.pdf
Injection Well Permit Reporting	AR-09 Displaced Brine 1-30-25.pdf
Injection Well Permit Reporting	December 2024 AR-09 Well Report.pdf
Injection Well Permit Reporting	January 2025 AR-09 Well Report.pdf
Isometric Certify Source Documents	See https://registry.isometric.com/account/certify/project/prj_1HZSSWBQM1S08H83/ghg-statement/ggs_1JQC60KVR1S0JS6Q
LCS Delivery	Univar 01092025.PDF
LCS Delivery	Univar 01282025.PDF
Misc. Documentation	[External] Charm + AECN Blended Batch Testing Management.docx
Misc. Documentation	Charm BO Validation FARs.xlsx
Misc. Documentation	Dec_Jan Verification Guidance.docx
PDD	Charm Kansas Bio-OIl pdd v1.8.pdf
PDD	V 1.9 Charm Industrial G.P. Bio Oil Sequestration PDD_Updated.docx
Vaulted emplacement data logs	185-9704.pdf
Vaulted emplacement data logs	185-9705.pdf
Vaulted emplacement data logs	185-9740.pdf
Vaulted emplacement data logs	185-9750.pdf
Vaulted emplacement data logs	185-9753.pdf
Vaulted emplacement data logs	185-9768.pdf
Vaulted emplacement data logs	185-9769.pdf
Vaulted emplacement data logs	185-9774.pdf
Vaulted emplacement data logs	185-9778.pdf
Vaulted emplacement data logs	185-9779.pdf
Vaulted emplacement data logs	185-9803.pdf
Vaulted emplacement data logs	185-9804.pdf
Vaulted emplacement data logs	185-9816.pdf
Vaulted emplacement data logs	185-9817.pdf
Vaulted emplacement data logs	185-9830.pdf
Vaulted emplacement data logs	185-9831.pdf
Vaulted emplacement data logs	185-9832.pdf
Vaulted emplacement data logs	185-9833.pdf
Vaulted emplacement data logs	185-9868.pdf
Vaulted emplacement data logs	185-9870.pdf
Vaulted emplacement data logs	185-9884.pdf
Vaulted emplacement data logs	185-9885.pdf
Vaulted Salt (brine) delivery ticket	1_21_25 Salt Delivery.pdf

Vaulted Salt (brine) delivery ticket	1_28_25 Salt Delivery.pdf
Waste Disposal	Bio-Oil Disposal EF Calc Sheet.xlsx
Waste Disposal	Bio-Oil Incineration.pdf
Waste Disposal	Bio-Oil Open Dump EF.pdf
Waste Disposal	Charm KS Dec_Jan RP Disposals Sheet.xlsx
Waste Disposal	Railcar Disposal Sheet - GATX 032955.xlsx
	2024_04_03_9376531.pdf
	2024_09_09 009376532.pdf
	2024_09_16 009376534.pdf
	2025_01_03 53776.pdf
	76350665619__63A06011-7264-4BC8-8888-5D8BD907CA31.HEIC
	76350687414__30D5D2C9-9F05-48CB-BC22-3D35B609A454.HEIC
	76350694883__27738CAD-DF98-4A97-AFD5-C70310935F24.HEIC
	76350698399__D110D82A-2A18-4832-891D-77B44146FB8D.HEIC
	76350702449__CE74715B-D706-4B0A-A8F1-6FC500718E1C.HEIC
	Biling doc 1874 CORRECTED PO NUMBER.pdf
	bill_252641473420 (1).pdf
	Billing Document 1870 CORRECTED PO and Period.pdf
	Billing Documents 1785.pdf
	Billing Documents 1788.pdf
	Billing Documents 1790.pdf
	Billing Documents 1792.pdf
	Billing Documents 1794.pdf
	Billing Documents 1796.pdf
	Billing Documents 1824.pdf
	Billing Documents 1826.pdf
	Billing documents 1831.pdf
	Billing Documents 1842.pdf
	Billing Documents 1844.pdf
	Billing Documents 1850 - corrected.pdf
	Billing Documents 1863 CORRECTED PO.pdf
	Billing Documents 1902.pdf
	fixedgas-gd10series-gd10p-brochure-english.pdf
	Load 1 Delivery Log.xlsx
	Screenshot 2024-08-28 at 9.04.43 PM.png
	Screenshot 2025-02-24 at 2.30.49 PM.png
	Screenshot 2025-02-24 at 2.37.57 PM.png
	Screenshot 2025-03-03 at 4.28.24 PM.png

APPENDIX 2: VERIFICATION AUDIT FINDINGS AND RESPONSES

ID No	Type	Finding / Issue	Required Action	Supplier Response	Conclusion / Resolution
	Charm Produced Bio-Oil Findings				
1	Clarification	Is any additional information provided by Altitude Forestry for each biomass delivery to support evidence and collection and use of their biomass? Agreement information was provided for Validation, and Charm logs indicate origin and destination, but no BOL is provided - only weigh tickets from local commercial scales	Submit additional information if available	Altitude is a small, family-run firm and does not provide official BOLs as part of their operations. The combination of the supply agreement, specifying harvest operating locations, harvesting timeline, product sepcifications, and quantity is meant to serve that purpose in combination with the scale tickets provided for each load.	Complete - prior validation evidence is adequate
2	Clarification	Biomass delivery weigh tickets typically provide only gross weight of load. Load 1 Delivery Log does not provide the vehicle weight for determination of net.	Please include gross, empty truck weight, and net in log for easy reference to weigh tickets	Several notes on this point: 1. Tare weight scale tickets have been added to the Weight Tickets folder 2. On closer review, the net weights provided by the transport provider was an overestimate of the cargo mass in multiple cases. This was a result of the provider rounding the tare weight. These discrepancies have been corrected. The log now shows the net weight as the difference between the gross weight and tare weight for each delivery for clarity	Complete. Data updated and clarified

3	Misstatement	Load 1 Delivery Log spreadsheet does not align with weigh tickets provided. For example: - 6/18/24 weigh ticket provided does not appear in log - 4/11/24 weigh ticket provided does not appear in log - 4/18/24 Log includes 2 entries but only one weigh ticket provided	Please review and update delivery log spreadsheet and documentation and calculatiosn that use the delivery logs	Entries have been updated -- A few of the ticket documents, including 6/18, contain two tickets in one file. Adjusting the delivery log totals to reflect both the adjustment in finding #3 and adding the tickets identified here also resulted in an adjustment to the overall transport delivery emissions allocation for the reporting period. This is now reflected in an updated version of the Reporting Period Summary document. The adjustments do not make a material impact on the carbon intensity of the pyrolysis products for the period. The small net increase in cargo mass after all adjustments is reflected in a slight increase in BCUs allocated to the period which, in turn, is reflected in the BCU tracker. The remaining emissions to allocate did not increase materially as a result of the changes. The increase of the end bio-oil CI from .4114 to .4115 represents a total of 2.463 kg CO2e difference in the production period.	Complete. Data updated and clarified. Non-material
4	Clarification	In Load 1 Delivery Log and Miniforge Biomass transport Allocation…Spreadsheets, what is the source and why is the 'fuel usage for loading and unloading' column included. It does not appear to be used, as emissions are calculated based on mass of biomass delivered and emissions factor based on biomass mass	Clarify	The "fuel usage for loading and unloading" (column k) is an estimate and is not used for MRV. The calculation used for MRV is reflected in the "Delivery Log Handling Emissions" cell (B10) of the calc sheet. That value is computed by multiplying the total tonnage delivered by the emission factor for log handling sourced from Auburn University. The column has been removed from the updated version of the delivery log	Complete.

5	Recommendation	Please include all formulas in spreadsheets for ease of review and tracing during verification. (i.e. PP1 Summary - Final Submission tab in Production Period 1 Summary file does not include formulas in Column C for majority of calculated values)	Update spreadsheets if possible to improve ease of traceability	Formulas for each parameter are in the individual calc sheets for that parameter. Because the formulas reference data that is in the sheets, values are hard-coded in the summary to avoid a reference error. This is a temporary solution as our tech team is working on integrating all datapoints in Ledger. In the meantime, any output for any parameter in the Summary can be verified by the calc sheet for that parameter	Complete
6	Recommendation	In the Production Period 1 Summary -- Final Submission file, Efs...tab, the emission factor for the log handling (Auburn source) is not included, nor is propane EF, nor other non transport factors. Recommend ensuring all emission factors, constants, etc. are documented consistently on this tab if used in the workbook. Provide a link to source for ease of reviewing	Update spreadsheets if possible to improve ease of traceability	No problem, working on this update	Not Required
7	Omission	The PPD referenced on the Isometric website (v1.8) does not appear to have the information regarding use of RECs and calculation approach included. Please update	Update PDD to include RECs	Will be updated by COB Wednesday -- Will send a revised version to Cassie to post	Complete - v 1.9 PDD provided
8	Recommendation	To improve stakeholder outreach and community engagement documentation, provide documentation of attendance lists for events as well as lists of questions and answers to demonstrate the level of engagement	Provide additional supporting information if possible	Thank you for the feedback -- This has been communicated to our Community Relations manager	Not Required
9	Recommendation	Provide an improved mechanism for community complaints or inquiries. For example - provide a specific link on the Contact Us or Community Benefits web pages for 'Inquiries or Complaints from the community' or similar	Provide additional supporting information if possible	Thank you for the feedback -- This has been communicated to our Community Relations manager	Not Required

10	Clarification	The 2024.11 Charm Industrial Waste Profile file contains lab reports with significant redactions. Can you clarify what was redacted? Was there additional data regarding non-related samples?	Clarify	The document was submitted for regulatory reporting -- The redacted content is limited to either 1) Commercial information that is not relevant to the regulator (e.g. AECN facility address), or 2) Data that is not relevant to regulatory reporting	Complete
11	Clarification	Batch 300001 is included in the file with Batch 200068. The MRV Ops log file '200068_300001 MRV Ops Notes' indicates only batch 200068. Is the Charm oil tab for batch 300008 and the other tab for 200068 (AECN oil)?	Clarify	200068 and 300001 are the same batch. Prior to introducing the Charm bio-oil mixed batches all batch numbers started with the number 2. Charm mixed batches started off the "3 series" of batch ID numbers starting with 3. Because this was the first batch, both the 2-series and 3-series numbers were used to prevent operator confusion. Now that both numbering conventions are solidified and in use this will not be an issue in the future	Complete
12	Misstatement	The Charm Bio-oil density and water content for Batch 300006 in 300006 MRV Ops Notes file do not match the SGS lab reports for 300006. Same for Batch 300003. This does not impact the ultimate C removal value, as the mass and carbon content are properly used. However, this could impact the salt addition and Cl- compliance requirement in the future (not impacted here)	Clarify	Density and water content for AECN oil are reflective of the COAs provided by AECN for each injection batch. The density measurement from the MRV tests is not appropriate to use for determining the quantity of salt to add because the sample represents finished injectate which already has salt and LCS added. The value of that parameter is only to show that the injectate is dense enough to sink in brine. The AECN result shows the density and WC of the raw oil, so it is appropriate to use to determine the levels of salt required to satisfy permit requirements. To help clarify, I've uploaded the COAs for all of the BOLs used to the AECN BOLs folder. I've also included a spreadsheet called "AECN COA Key" that shows which BOLs fall under which COAs	Complete

13	Clarification	On 'Bio-Oil Transport Calculations' tab of 'Production Period 1 Summary - Final Subission' file, where does the 'Quantity of Oil Delivererd (kg) = 15828 kg derive from? It might possibly be from the AECN oil percentage times the amount of injectate delivered?	Clarify	15828 is the total net weight of the bio-oil delivered during the first Charm bio-oil deivery to KS. That total is noted in the Charm Deliveries Truck Mass Allocation Calculator, cell D30 -- Noting the 1kg discrepancy, I have updated the value in the PP summary to reflect 15827 kg net weight of oil delivered	Complete
14	Misstatement	On 'Batch Details' tab of 'Production Period 1 Summary - Final Subission' file, The Charm Bio-oil value does not match the value for 300001 indicated in the MRV Ops notes for 300001 (2695 vs2670 kg). The Batch carbon content listed is only the C content for AECN oil, the blended oil content is 42.56%. Please clarify or correct	Clarify & correct if needed	Corrected, thank you for the catch!	Complete
15	Clarification	Please clarify the Vaulted Electric meter recording calculations. For example, for 185-9740 file (Batch 300001), the readings indicate 7997 -7993 = 4 *40 = 160 (not 80) and 5711-5705 = 6 (not 3). These are used elsewhere, including PP1 Summary calcs for Charm oil CI	Clarify & correct if needed	I reached out to Vaulted on this and they confirmed that for many of Charm's injections are "dual emplacement" events where Vaulted is injecting their own material while Charm is injecting into AR-9. In those instances, they divide the total meter consumption in half to allocate between Charm and Vaulted injections	Complete
16	Clarification	Is there any additional documentation of the starting and ending linear feet of wood during the reporting period?	Clarify	No -- Measurements are taken manually by site staff who physically measure the piles to get the value. Because the measurements are done manually, the only data recorded is in the form of communications from the staff taking the measurements	Complete
17	Correction - Edit	On 'PP1 Summary -- No PRs' tab of 'Production Period 1 Summary - Final Submission' file, it says 'Total Biomass Consumed (kg) (Uncertainty-Adjusted)' but this is not uncertainty adjusted (and shouldn't be). Please relabel.	Edit	Thank you for catching that typo -- It has been corrected	Complete

18	Recommendation	The emissions from transporting waste for disposal are likely significantly overestimated based on use of the full truck scale measurement error added to the mass of Charm's waste. It is highly unlikely that Charm's product is the only waste on the truck and the error could be allocated as a percentage of the total load instead of fully to Charm's single drum	Update if desired	That is a helpful note, thank you!	Not Required
19	Recommendation	To the BCU and RECs Tracker sheets, please add a column to the BCU/REC inventory indicating how much total is used and remaining from each BCU	Modify tracking sheet	Updated	Complete
20	Misstatement	BCU tracker indicates 19.98 MTCO2e BCUs used for Charm Bio-Oil Production. This covers the Diesel use only in the PP1 Summary spreadsheet. Sheet indicates that 4.11 + 0.25 MT CO2e BCU were used for waste disposal and biomass feedstock handling and delivery associated with Charm BO production	Clarify & correct if needed	The total has been updated to 24.67 MT CO2e to cover all production parameters to which BCUs have been applied, including the adjustment to transport emissions allocation noted above. In addition, a "Procured Renewables Totals" section has been added to the Production Period Summary for clarity	Complete
21	Omission	Please provide a copy of the contract for REC purchases (retirement certs provided but not contract as required by EC1)	Provide additional supporting information (contract)	The contract for REC purchases has been added to the verification materials folder in the subfolder "Additional Verification Documents"	Complete

22	Forward Action Request	Please provide evidence and accounting of disposal of oil from lots 200062 and 200063 when disposed of in future		The site disposals tracker (Charm KS Dec_Jan RP Disposals Sheet)documents the disposal of waste materials from the site, including waste bio-oil. This reporting is separate from reporting the upstream emissions of disposed batches, which is batch-specific. Bio-oil disposed on site includes both oil from these disposed batches and other bio-oil waste from the extended R+D period in 2024. For that reason, the bio-oil source related to a particular disposal is not distinguishable. However, disposal emissions from all bio-oil disposals are captured in the tracker, and the upstream emissions from all disposed batches are captured in their own calc sheets. Given the above, it is assured that all emission related to any disposed batch are accounted for. This is an especially conservative approach since it also attributes all disposal emissions due to the disposal of waste bio-oil from the R+D period to the Reporting Period during which the disposal occurs.	Complete
23	Misstatement	Date of service entry for CO REC solar facility is misstated on REC tracking sheet (Charm BCU Verification Inventory tracker). Cell E4 states "2 months elapsed between the CO generator entering service (11/1/21) and the initiation of the project (3/1/24)" while the REC project evidence indicates a start date of 12-29-2023. This does not impact REC eligibility	Edit	Updated, thank you for the catch!	Complete
24	Omission	Please provide the KS electricity emissions factor for RECs facility similar to what was provided for CO ('CO Grid Emissions Calc Sheet')	Provide additional supporting information	The calc sheet has been added to the "Additional Verification Documents" folder in the verification package	Complete
25	Forward Action Request	Please provide a copy of the October 24 emissions control burner test when used		Confirmed, that will be included in the submission the subsequent Production Periods	FAR

		in future removals accounting for tail gas emissions			
26	Clarification / Misstatement	Emission 3196 on Lot 1082 (transport from AECN to Charm) - the mass indicated on Isometric Certify is 13106 kg (matches the MRV Ops workbook), but the BOL from AECN and all source data on Certify indicate 20.26 / 20.57 tonne (depending on which value you use). And, lot and transfer graph indicates 100% of that lot eventually is included in Removal 755	Clarify & correct if needed	Emission 3196 on Lot 1082 originates from BOL 1790. The remaining ~7463 kg not disposed Removal 755, was in Batch 2-65. That was the batch which was injected at Vaulted but not sampled. Since that left us with no way to quantify the C content, it was treated as a disposal. The upstream emissions for that portion of the BOL are reflected in the Site Emissions 376 and 377 as part of the total oil included in the batch. That is confirmed by the Ops Notes for Batch 2-65. To make the adjustment "work" on Ledger, I had to adjust the total mass for BOL 1790 as an input to Removal 755/Batch 2-66 to just reflect the amount used in the batch rather than the full BOL. Otherwise, I would have "disposed" the remainder of the BOL, which would push the remaining upstream emissions onto 755. Since they are already accounted for as a Site Emission, that would have effectively double-counted the upstream emissions from the ~7463 kg that went into Batch 2-65. So, all of the emissions for BOL 1790 are accounted for, but it makes sense that it wouldn't be clear where the rest of the oil delivered went.	Complete
27	Clarification	Confirming that the 'Generic Diesel / Non-Deisel Emission at Site 34 (Ajs)' included in removal 755 and 761 are all emissions assocaited with the batches of oil that were disposed	Clarify	All of these can be found in the "Disposed Oil" sub-folder of the Waste Disposal folder in Site Emissions. Only one railcar was disposed of, but two batches (2-62 and 2-63) were scrapped. 2-65 was injected w/ no sample for C content (see question 1). By emission SE 386 is the non-diesel upstream emissions from (scrapped) batch 2-63 SE 385 is the diesel upstream emissions from (scrapped) batch 2-63 SE 383 is the diesel upstream emissions from the disposal of the contents of the railcar GATX 032955	Complete

28	Clarification / Misstatement	BCU tracker shows 260 MT of BCU CO2e used. GHG statement shows 236MT. The 236 appears to not include the additional 24.67 MT CO2e for Charm Bio-oil production	Clarify & correct if needed	BCUs Yes, that is a good point. The GHG Statement Results reflect the totals given in Isometric DMRV. The issue is that the 24.67 MT CO23 in BCUs applied to the Charm bio-oil production emissions are already reflected in the total by the carbon intensity of the Charm oil. Because that is calculated externally, they don't show up in the DMRV reductions total. However, if I add them to the results, that would make the total removal incorrect since they would effectively have been double-applied. RECs We submitted this GHG Statement before all of the component blueprints were ready to go For RECs. Because 100% of the kWh consumed are matched with RECs, the emission factor used for electricity consumption corresponds to the emission factor for the RECs rather than the emission factor for the residual grid mix. So, the emission effect is there, but it does not fall into the reductions bucket. In general, I would be happy to add a note to the temporary changes section in the GHG Statement noting these, so it's clear a) What the total usage of RECs and BCUs is and, b) Where it shows up in the reporting	Complete - GHG Statement updated and clarified
29	Omission	A copy of the revised UIC permit for the use of AR-09 well is not provided	Provide additional supporting information	Current permit has not been updated by KDHE, but covers additional well openings	Complete. All permit actions for AR09 cavern opening and well installation completed and properly documented. KDHE may or may not issue updated permit with well specifically identified, but does not affect permit

					approval of new cavern.
30	Forward Action Request	Validation reports indicated that AECN was going to be performing pyrolyzer stack emissions testing. Has this been completed yet?	Please provide AECN testing report. Note, likely not applicable for this reporting period, but should be updated when possible		Testing not completed yet. Remains as FAR
31	Forward Action Request (Prior)	Provide initial LCA for Charm-produced bio-oil demonstrating net negative emissions and documenting LCA data usage as required by protocol		Cradle-to-grave LCA is included in the Production Period 1 MRV summary. The first injection of Charm bio-oil is modeled from production through injection	Complete
32	Forward Action Request (Prior)	Provide uncertainty and sensitivity analysis for GHG Statement for production and utilization of Charm bio-oil		A sensitivity and uncertainty analysis is included in the Production Period MRV summary, using the LCA as an input for the sensitivity analysis. Uncertainty modifications are also noted in the Production Period Summary for each parameter modified	Complete
33	Forward Action Request (Prior)	Provide analysis in compliance with the requirements of protocol (Bio-oil testing results using USEPA RCRA SW0846 test methods) to demonstrate non-hazardous claim		A full characterization of bio-oil as an injectable material stream, including TCLP testing and SDS, was provided to and approved by KDHE. The documentation submitted is included for verification	Complete
34	Forward Action Request (Prior)	Modify PDD to include biomass storage site in Ft. Lupton operated by Charm		Completed prior to verification	Complete
35	Forward Action Request (Prior)	Modify PDD to reflect proper Crediting Period starting March 1, 2024 and continuing for 5 years		Completed prior to verification	Complete
36	Forward Action Request (Prior)	Verify bio-oil sampling completed in accordance with requirements of Monitoring Plans as well as UIC permit in the first verification of Charm bio-oil injection		Injections using Charm bio-oil are subject to the same set of tests as already-verified third party injections (% C content, density, TAN, pH, temperature, chloride levels) and documentation is included for each injection batch	Complete
37	Forward Action Request (Prior)	Demonstrate all aspects of LCA related to Charm-produced bio-oil and have verified		Cradle-to-grave LCA is included in the Production Period 1 MRV summary. The first injection of Charm bio-oil is modeled from production through injection	Complete

38	Forward Action Request (Prior)	Monitor requirement for submitting air permitting notification (APEN) in future verifications		Charm's CO facility is operating below the APEN threshold. Engineering resources are monitoring relevant emissions on an ongoing basis based on the results of emission testing scaled by throughput and operating hours.	Continue to monitor
39	Forward Action Request (Prior)	Review and validate permitting requirements and compliance status at future verifications		Site work is ongoing to complete the rainwater detention system required by the City of Ft. Lupton. The site remains in compliance with all relevant facilities permits.	Continue to monitor
40	Forward Action Request (Prior)	Update community engagement records to document planned activity. Include in future verifications		An updated report on community engagement activities is included in the documentation package for this verification	Continue to monitor
41	Forward Action Request (Prior)	Provide updated logs of spills, leaks, and process upsets for first verification of Charm-produced bio-oil		The only major process upset in the first Production Period was a bio-oil spill due to faulty tote handling. This resulted in the removal and disposal of bio-oil contaminated gravel which is accounted for in the Waste Disposal data for the production period	Complete - records of spills, upsets, and waste disposal monitored and accounted for
42	Forward Action Request (Prior)	Review site operations and document all changes from operations observed during site visit. Verify changes are accounted for in LCA, including additional equipment inventory and embodied emissions (equipment and construction)		Additional pyrolyzers and related ancillary equipment (e.g. work platforms, permanent fuel tanks for propane and diesel, shipping containers for human-machine-interface controls) have been installed and added to the equipment inventory with embodied emissions quantified by LCA. Ongoing sitework projects will be included as completed, but were ongoing at the close of the Production Period	Complete - records of new equipment are maintained and embodied emissions accounted for

43	Forward Action Request (Prior)	Provide new emissions testing report and methane emissions calculation documentation. Account for new tailgas emissions values in LCA and have verified in first verification for Charm bio-oil after changeover to low-swirl burner		The installation of a low-swirl burner has not moved forward. Instead, the engineering team made modifications to the existing combustion infrastructure to improve performance such as using propane to increase system temperature and improved temperature monitoring. As a result of these changes a second emissions test was commissioned in October 2024 demonstrating improved performance. Because the rollout of the improvements was progressive over the course of the production period, the effect was uneven. To ensure a conservative estimate of removals, the initial test showing higher emissions per kg of biomass consumed, which was included in the project validation documents, is used for all of the production period. In subsequent periods, the more recent emission testing will be used, reflecting the impact of these improvements	Provide new emissions data for next Reporting Period
44	Forward Action Request (Prior)	Provide all required monitoring records for bio-oil analysis prior to verification using Charm bio-oil		Requirements for this are addressed in FARs 3 and 6	Complete
45	Forward Action Request (Prior)	Provide specifications, design, installation information, calibrations, and operational data for the new gas monitoring equipment at Vaulted		Vaulted has installed a GD10P Infrared Point Gas Detector at the AR-09 wellhead. Equipment specs have been included as additional documentation. The detector is monitored by Vaulted, who have provided a written and signed commitment to alert Charm Industrial within 2 hours of a verified gas detection	Provide evidence of installation and usage

APPENDIX 3: VERIFIER QUALIFICATIONS

Supporting documentation, including verifier resumes, and verifier or corporate accreditations are also included in this appendix.

Verifier Qualifications	Criteria Met?	Evidence / Notes *(note how the criteria was met, specific documents - resume/CV, publications, certifications, etc.).*
Verifier has relevant technical knowledge of the type of technology being evaluated and carbon removal processes in general		
A) Does Verifier have:		
1. An in-depth technical knowledge of the technology type under verification;	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 Environmental Technology Verification (ETV) as a Type A (third party) Inspection Body (ANAB Certificate Number: AI-2618). The technical scope of 350's accreditation includes verification of performance and environmental impact as it relates to design, materials, equipment, installation and operations of technologies in the categories of Energy, Clean Production and Process, and Air Pollution Monitoring and Abatement. As documented in 350Solutions' ETV Standard Operating Procedure (ETV QPM 350-223-03), and Quality Systems Procedures for verifier qualifications (QSP-350-005-02), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to verifier qualifications and procedures. These procedures and quality management programs are generally relevant to verification under the Isometric Standard. Note that verifications completed for Isometric are not equivalent to ISO 14034 verifications. 350 staff have participated in the evaluation and verification of novel technologies that sequester carbon via various methods, including biomass conversion to liquids, solids, and other products which are then permanantly stored in ways such as land application or geologic storage, conversion of captured CO2 into building materials and co-products, and the production of chemicals, fuels, and products via biomass pyrolysis and gasification. 350 also served as lead verifier for the Carbon XPrize competition and contributed to the development of procedures and processes for verification of relevant calculations, modeling, and statistical methods in order to assess team results and calculations of performance metrics and uncertainty. 350 has demonstrated knowledge of data quality and data validation approaches and execution in supporting verification of performance claims and results.
2. Knowledge of specific risk areas associated with performance of such technologies (i.e. common failure points, performance issues, barriers to scaleup);	☑	
3. Knowledge of the environmental implications related to the use of the technology from a life cycle perspective, such as impact of the technology on lifecycle CO2 emissions and carbon removal;	☑	
4. Knowledge of relevant applicable test methods and standards for evaluating performance or impact of the technology;	☑	
5. Knowledge of relevant calculation, modeling, and statistical methods in order to assess test results and calculations of performance metrics and uncertainty, as applicable;	☑	
6. Knowledge of data quality and data validation approaches, including QA/QC procedures, for example.	☑	
Verifier is a credible independent 3rd party		
B) Is Verifier:		
1. third-party body independent of the team registered for the Isometric Registry	☑	350Solutions is accredited to ISO/IEC 17020:2012 and ISO 14034 ETV as a Type A (third party) Inspection Body. As documented in 350Solutions ETV Policy Manual (ETV QPM 350-200-03), 350Solutions conforms to the requirements of ISO 17020 Annex A with respect to impartiality for Type A inspections, pursuant to ISO 14034 activities.
2. Not directly involved in the design, manufacture or construction, marketing, installation, use or maintenance of the specific technologies submitted to Isometric for verification, or represent the parties engaged in those activities.	☑	
3. Not part of a legal entity that is engaged in design, manufacture, supply, installation, purchase, ownership, use or maintenance of the items inspected.	☑	

Tim Hansen, P.E.
Founder and CEO, 350Solutions

EDUCATION:
B.S., Chemical Engineering, University of Virginia, 1993
M.S., Engineering Science, Thayer School of Engineering, Dartmouth College, 1995

EXPERIENCE SUMMARY:
Mr. Hansen has 26 years of experience in management of energy and environmental technology development and demonstration projects and programs, as well as multimedia environmental engineering efforts. These majority of his recent work has focused on the evaluation of innovative carbon capture, utilization, and removal technologies. Mr. Hansen has led the development and management of large technology evaluation programs in the advanced energy, transportation, and climate change areas.

RESEARCH AND PROFESSIONAL EXPERIENCE:
2019-Present Founder – CEO, 350Solutions, Inc.
Owns and operates a small cleantech engineering consulting business focused on the independent evaluation of new cleantech innovations and their impact on the environment and carbon emissions. Provides engineering consulting, testing and evaluation, techno-economic assessment, and other support to companies developing, using, or investing in new clean technology innovations. Manages administrative, business development, and project activities for 350Solutions.

2012-2019: Director - Energy and Environment, Southern Research
Manages scientific and technical staff performing research, development, and evaluation of innovative clean energy technologies. Projects range from $25,000 to $6million in size, and are funded by the US Department of Energy, Department of Defense, and commercial partners. Technical focus areas are conversion of biomass to fuels and chemicals, carbon capture and utilization, energy efficient building technologies and renewable energy generation.

2009-2012: Program Manager – Transportation & Climate Change Technology, Southern Research
2003-2009 Sr. Project Leader, Environmental Engineer, Southern Research
1996-2003 Environmental Engineer, Bensinger & Garrison Environmental

PROJECT EXPERIENCE:
Mr. Hansen has executed several independent technology performance verifications of emerging carbon, energy and transportation technologies, as CEO of 350Solutions, Director of Energy & Environment at Southern Research, and Director of the U.S. EPA's Greenhouse Gas Technology Center. Mr. Hansen has completed clean technology evaluations for the Department of Defense, state energy agencies, commercial clients, investors, and technology developers, involving evaluation of commercial feasibility, economic and environmental impacts, and technology performance. Mr. Hansen served as the Measurement and Verification Program Lead for the NRG COSIA Carbon XPrize – a $20M prize competition for technologies that capture and beneficially utilize CO_2. Mr. Hansen also served as U.S. Technical Expert for the development and implementation of ISO 14034 – Environmental Technology Verification, an international standard, issued in 2016

Lily Schacht
Carbon Removal Verification Engineer, 350Solutions

EDUCATION:
MS, Environmental Engineering, University of Wisconsin – Madison, 2019
BS, Chemical, Energy, and Environmental Engineering, Washington University in St. Louis, 2017

EXPERIENCE SUMMARY:
Lily Schacht is an Environmental and Chemical Engineer with experience in process engineering, environmental chemistry, analytical methods, and life cycle analysis (LCA). At 350Solutions, Lily works on verifying carbon dioxide removal (CDR) technologies, with a focus on mineralization-based pathways, including enhanced weathering, direct air capture, and ocean alkalinity enhancement. Previously, Lily led agronomic research at an enhanced weathering CDR supplier where she organized field trials across multiple states to quantify carbon removal and crop yield changes after rock application. Before that, Lily developed rapid prototyping instrumentation to optimize a biomineralization process in concrete production and aided in scaling up the process to pilot-scale. Lily also built environmental impact models for process variable sensitivity analysis of demo-scale manufacturing processes. These models were used to drive the direction of research and development to minimize product life cycle impacts. Throughout her career, Lily has developed analytical chemistry methodologies for both liquid- and solid-phase analyses.

RESEARCH AND PROFESSIONAL EXPERIENCE:
March 2024 – Present: Carbon Removal Verification Engineer, 350Solutions
Verify CDR technologies on behalf of registries and the XPRIZE Carbon Removal challenge. Specializes in mineralization pathways, including mineralization kinetics, measurement methods, and open-system modeling.

Nov 2022 – Feb 2024: Researcher, Lithos Carbon
Quantified carbon removal rates and agronomic impacts of enhanced weathering on cropland across six US states; Evaluated chemical analysis methods for precision relative to cost.

Nov 2023 – Jan 2024: Independent Consultant, Keel Labs
Built an environmental impact model to evaluate potential material and process changes; Recommended areas for reducing material usage up to 80%

Dec 2020 – Oct 2022: Research Scientist II
Guided experimentation and data analysis throughout all R&D teams to inform techno-economic analysis (TEA) and LCA; built and led the carbonate biomineralization prototyping workstream for rapid iteration; developed real-time measurement techniques for critical process parameters in solid state

SELECTED PUBLICATIONS & PRESENTATIONS:

- **Schacht, L.**, Baum, M., Liu, H., & Yap, M. (2023) Scaling Enhanced Rock Weathering: Agronomic Impacts at Field-Scale [Abstract]. ASA, CSSA, SSSA International Annual Meeting, St. Louis, MO.
- **Schacht, L.** and Ginder-Vogel, M. Arsenite Depletion by Manganese Oxides: A Case Study on the Limitations of Observed First Order Rate Constants. *Soil Syst.* 2018, 2(3), 39. https://doi.org/10.3390/soilsystems2030039